PROJECT BRIEF / ML SYSTEM DESIGN & AGENTOPS

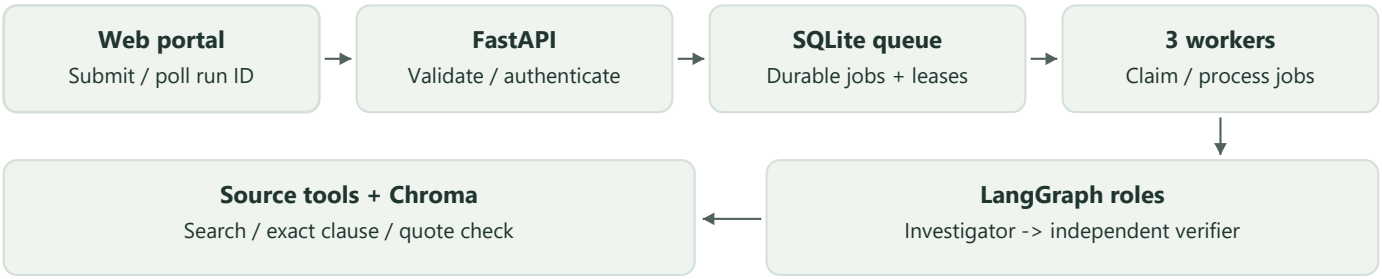
Academic Regulation & Student Petition Triage Service

Mini Project 15 | UE24AM342AA5 | Human-reviewed academic support

01 WHAT WE ARE BUILDING

A source-grounded service that turns a fictional academic petition into a cited, conditional recommendation for an academic officer. It identifies applicable rules, required supporting records and exceptions, then recommends processing under a provision, referral, or a clearly supported restriction. The officer always makes the final decision.

02 ARCHITECTURE AND REQUEST FLOW



Roles select source tools; workers enforce the final source gate and attach a conditional position. The stored brief is returned through the API for human review.

03 KEY TECHNICAL DETAILS

Retrieval LangChain + local all-MiniLM-L6-v2 embeddings; 180 handbook pages, 509 records and 13 reviewed clauses. Hybrid search, full quotations, page/section metadata and incremental re-embedding.

Agent execution Two distinct prompted roles; three selectable source tools; typed outputs via Pydantic. Up to six model turns/tools per role and one re-search; unsupported output escalates or falls back.

Reliability & access Workspace-key login, quarantine and input/output safeguards; Redis cache/rate limits; timeouts, bounded retries, circuit breaker and isolated deterministic fallback.

Deployment & visibility Docker Compose with API, workers, indexer, separate stub model service and Redis. OpenTelemetry/Jaeger traces, Prometheus metrics and Grafana dashboards track runs and role usage.

04 INTENDED END-PRODUCT CAPABILITIES

The end product should let a user submit a petition, track its progress and receive a clear review brief with applicable rules, exact citations, missing evidence and a conditional recommendation. Officers should be able to inspect the sources, review exceptions and record feedback, while the service handles concurrent requests and exposes traceable failures.

Scope: Development uses simulation with real retrieval to limit hosted usage. Source fidelity is not general semantic proof. The intended product supports an officer rather than making autonomous academic decisions; all facts and recommendations require human review. Use synthetic petitions only.